



ASSIGNMENT No. 3 Problem Based Learning

Course Title: Computer Organization and Assembly Language

Course Code: CEN-323

Class: BS (CS)-4(A/B)

Course Instructor: Dr. Engr. Ghulam Muhammad

Submission deadline: 25-04-2025

Marks: 10

Instructions

- This assignment is a problem-based learning assignment which can be performed individually or in a group of 2 students
- Deadline will not be extended
- Copied assignments will be marked zero
- Prepare a **pdf file** and upload to LMS

1. Problem

Imagine you are part of a team of CPU performance architects at NextGen Pro Limited and tasked with resolving an issue with the current instruction set architecture (ISA) designed for a next-generation processor. The ISA designed by another team of architects at your company has inefficient instruction encoding that leads to less functionalities consuming more number of bits in an instruction as follows in the example:

14-bit	12-bit	16-bit	4-bit	4-bit	8-bit	6-bit
Opcode	Operand 1	Operand 2	Addressing mode for operand 1	Addressing mode for operand 2	Reserved Byte	Miscellaneous

64-bit Instruction Format

Apply your knowledge of instruction set design to create an ISA that balances performance and efficient instruction encoding. **Develop** the format for each instruction, specifying the opcode, operand fields and addressing modes in the best possible way to achieve efficient instruction encoding for the following operations specifically as you have learned in this course. You may change the size of each field as per your requirement. Reserved Byte and Miscellaneous field will remain constant. Also specify which addressing mode you are using for each instruction and show its assembly language equivalent instruction.

Arithmetic Operations:

Addition (ADD)

Multiplication (MUL)

Logical Operations:

AND

XOR



Comparison Operations:

Equal (EQ)

Not Equal (NE)

Rotate Operations:

Rotate Right (ROTR)

Data Movement Operations:

Move (MOV)

Control Flow Operations:

Branch (BR)

Jump (JMP)

2. Targeted CLOs

CLO2, PLO2, C3

3. Evaluation Criteria

- **Instruction Encoding** 3 marks
- **Solution through encoding efficiency** 3 marks
- **Viva** 4 marks